Polynomial-depth learning for ReLU and GELU NNBayes stable under neuron merging

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Abstract

We prove a sample bound polynomial jointly in Boolean teacher width and depth for Bayesian ReLU and GELU networks, at every learner width above the compiled teacher width and every mean-field-or-stronger Gaussian scale. The prior retains networks whose clipped predictions survive averaging neurons into a limited number of populations. Populations may have arbitrary sizes; their permitted number is calibrated by scale-normalized readout energy. Every layered bounded-fan-in Boolean circuit has a positive-mass implementation satisfying the condition. ReLU perturbations need not contract through depth: an exponentially small parameter radius has only a polynomial negative-log-prior cost. Positive homogeneity removes population-mass parameters from the ReLU computation; a uniformly Lipschitz perspective transformation controls them for GELU. The bound is quadratic in teacher width up to logarithms, with a coarser literal quadratic-width envelope of degree nine in the remaining resources. This is a structural prior restriction, not a proved property of ordinary internally stable or gradient-trained networks. No Lean verification is claimed.

1 The learning question and the restriction

For ReLU networks, product-of-layer-norm capacity bounds give a width-independent Gaussian Bayesian learning theorem at fixed depth, but its degree in teacher width grows with depth. We obtain polynomial depth dependence by restricting the prior to networks whose predictions are stable under merging redundant neurons. This supplies a direct finite description with polynomially many coefficients.

Two features distinguish the argument from bounded-activation population constructions. ReLU's positive homogeneity removes population sizes from the normalized computation. And although perturbations can amplify exponentially through depth, their admissible parameter radius enters the prior-mass estimate only logarithmically. The result concerns exact Bayesian conditioning, not an efficient posterior sampler or the output law of gradient descent.

2 Model and population condition

Fix $d \geq 1$, $L \geq 2$, $H = L - 1$, and $\mathcal{X} = \{-1, 1\}^d$. Write $\bar{x} = (x, 1)$. All hidden layers have width N:

$$h^1 = \mathrm{ReLU}(W^1 \bar{x}), \quad h^s = \mathrm{ReLU}(W^s h^{s-1}) \ (2 \leq s \leq H), \quad z_w = v^\top h^H.$$

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There are no separate internal biases; counted constant wires can supply offsets. The raw prior $\mu_{N,t}$ has independent Gaussian entries of variances

$$t^2/d \text{ in } W^1, \qquad t^2/N \text{ in } W^s \ (s \geq 2), \qquad t^2/N^2 \text{ in } v, \quad 0 < t \leq 1.$$

Thus $t = N^{(1-\gamma)/2}$ includes every $\gamma \geq 1$. Set

$$E_v = N \|v\|_2^2, \quad M(w) = \max_x |z_w(x)|, \quad T(u) = \min\{1/2, \max\{-1/2, u\}\}.$$

At layer s, partition the neurons into q_s nonempty populations, of arbitrary sizes $k_{s,a}$. Let column a of F_s be the normalized indicator of its population, with entries $1/\sqrt{k_{s,a}}$. Thus $F_s^\top F_s = I$. Put $P_s = F_s F_s^\top$ and define merged parameters

$$\bar{W}^1 = P_1 W^1, \quad \bar{W}^s = P_s W^s P_{s-1}, \quad \bar{v} = P_H v.$$

With $q = \max_s q_s$, the event $\mathcal{A}_N$ requires some such partitions satisfying

$$(q-1)M(w)^2 \leq 4E_v(w), \qquad \max_x |T(z_w(x)) - T(z_{\bar{w}}(x))| \leq 1/16. \tag{1}$$

The condition does not use the target or its complexity. The count budget is invariant under readout rescaling; the clipped comparison retains fixed margins. The complete event is invariant under neuron permutations. Populations are unsigned: arbitrary hidden sign flips are not symmetries of ReLU networks. Finitely many witnesses at fixed N make the event Borel.

The prior is $\mu_{N,t}$ conditioned on $\mathcal{A}_N$. One may additionally impose the original all-input margin event $|z_w(x)| > 1/2$ everywhere. Training further conditions on $f(X_i)z_w(X_i) > 1/2$. The theorem holds for either prior convention and tests the event $f(X)z_w(X) \leq 1/4$.

3 Explicit learning theorem

The target is a single-output layered Boolean circuit of width $C \geq 1$ and H stages, using NOR, COPY, and NOT. Constant wires count toward C; compilation from another bounded-fan-in basis must count its width and depth overhead. Bits are $0, 1$; target labels are $-1, 1$.

Define

$$\begin{aligned}
&a = 4, && b_L = \sqrt{16\log 9 + 8\log(4L)}, && S_0 = 2 + 2\sqrt{10C} + b_L,\\
&D_0 = d + 1 + H\sqrt{C}, && \rho = \min\left\{\frac{1}{16}, \frac{1}{64a\sqrt{10C}\, D_0 S_0^H}\right\},\\
&q_0 = d + 2 + (H-1)C, && B_0 = d + 10(H-1)C + 10a^2C,\\
&D_* = B_0 + q_0 \log\frac{e\sqrt{2\pi}}{2\rho} + \log 2.
\end{aligned} \tag{2}$$

For $0 < \varepsilon, \delta < 1$, set

$$\begin{aligned}
&B = \max\{8, 8[D_* + \log(4/\varepsilon)]\}, && Q = \lceil 1 + 16B \rceil,\\
&J = 4\left[D_* + \log(4/\varepsilon) + 2H\log Q + \frac{\log 2}{2}\{d + 2 + (H-1)Q\}\right],\\
&R = 1 + \sqrt{J}, && A_g = (d + 1 + Q)(R+1),\\
&\zeta = [32(H+1)A_g^{H+1}]^{-1}, && P_Q = Q(d+2) + (H-1)Q^2.
\end{aligned} \tag{3}$$

Theorem 1. *If*

$$n \geq \frac{2}{\varepsilon}\left[P_Q \log(3 + 4R/\zeta) + \log(1/\delta)\right], \tag{4}$$

then, with probability at least $1 - \delta$ *over iid inputs from any fixed distribution on* $\mathcal{X}$*, the posterior obeys*

$$\mathbb{E}_w \mathbb{P}_X\{f(X)z_w(X) \leq 1/4\} \leq \varepsilon$$

simultaneously for every $N \geq C$ *and* $0 < t \leq 1$*. All prior and posterior normalizers are positive. The same conclusion holds with the additional global-margin restriction.*

At fixed accuracy and confidence, a useful summary is

$$n = \widetilde{O}(LdQ + L^2Q^2), \qquad Q = \widetilde{O}(Ld + L^2C).$$

More generally the leading $1/\varepsilon$ and confidence term are explicit, and Q additionally depends logarithmically on $1/\varepsilon$. All depth dependence is polynomial. At fixed other parameters the width dependence is $O(C^2 \log^3(2 + C))$.

Corollary 2. *For a universal constant* A *and* $S = 2 + d + L + \log(4/\varepsilon)$*, a sufficient bound for these single-output bounded-fan-in teachers is*

$$n \geq \frac{A}{\varepsilon}\left[C^2 S^9 + \log(1/\delta)\right].$$

This coarser estimate prunes irrelevant circuit wires before absorbing logarithmic width terms into depth. Neither estimate is a claim of optimality.

4 A positive-mass Boolean implementation

Lemma 3 (Target cylinder)**.** *For every* $N \geq C$ *and* $0 < t \leq 1$*, a set of raw probability at least* $\exp(-ND_*/t^2)$ *lies in* $\mathcal{A}_N$ *and satisfies* $f(x)z_w(x) > 1$ *everywhere. It also satisfies the optional global margin.*

Proof. Partition the neurons into C balanced clone groups of sizes k_b, so $C/2 \leq N/k_b \leq 2C$. Implement gates by

$$\mathrm{NOR}(u, v) = \mathrm{ReLU}(1 - u - v), \quad \mathrm{COPY}(u) = \mathrm{ReLU}(u), \quad \mathrm{NOT}(u) = \mathrm{ReLU}(1 - u).$$

Propagate a constant-one wire by a copy. Substitute input bits $(1 + x_i)/2$ into first-layer gates. Each first-layer row has squared norm at most one. Each later logical row has sum of squared coefficients at most five, including repeated gate inputs. Divide each source coefficient equally among that source's clones. The teacher states $b^s(x)$ are exact $0, 1$ bits, constant on populations, with norm at most $\sqrt{N}$. Read out $z_* = a(2b^H_{\mathrm{out}} - 1) = af$, using the final constant wire. Combine coefficients if the output and constant wires coincide.

Let P average within these fixed clone groups. The teacher satisfies

$$PW^1_* = W^1_*, \quad PW^s_*P = W^s_*, \quad Pv_* = v_*.$$

The coefficient bounds and clone sizes give

$$\left\|W^1_*\right\|^2_F \leq N, \quad \|W^s_*\|^2_F \leq 10C, \quad a^2C/2 \leq N\|v_*\|^2_2 \leq 10a^2C.$$

The lower readout bound remains valid when output and constant wires coincide.

In raw coordinates $\sqrt{d}W^1$, $\sqrt{N}W^s$, and Nv, constrain each first-layer coordinate, each internal row's coordinate along every normalized clone-group direction, and each readout coordinate to an interval of radius ρt around its teacher value. There are Nq_0 independent constrained coordinates, with sum of squared centers at most NB_0. Internally

$$W^s = W^s_* + R^s + E^s, \quad R^s = R^s P, \quad \|R^s\|_{\mathrm{op}} \le \rho t\sqrt{C}, \quad E^s P = 0.$$

The remaining Gaussian bulk is independent of these coordinates. Two 1/4-nets of size at most 9^N give

$$\mathbb{P}\{\|E^s\|_{\mathrm{op}} > b_L t\} \le 2\exp(2N\log 9 - Nb_L^2/8) \le 1/(2L).$$

Thus all bulk bounds hold conditionally with probability at least 1/2.

Set $e_s = \|h^s - b^s\|_2/\sqrt{N}$. ReLU is 1-Lipschitz, giving $e_1 \le \rho(d+1)$. Since $E^s b^{s-1} = 0$ and $\|b^{s-1}\|_2/\sqrt{N} \le 1$,

$$e_s \le (\sqrt{10C} + \rho\sqrt{C} + b_L)e_{s-1} + \rho\sqrt{C} \le S_0 e_{s-1} + \rho\sqrt{C}.$$

Hence $e_s \le \rho D_0 S_0^{s-1}$ and $e_H \le 1/(64a\sqrt{10C})$. Also $\sqrt{N}\,\|v - v_*\|_2 \le \rho$, so

$$|z_w - z_*| \le a\sqrt{10C}\, e_H + \rho(1 + e_H) < 1/8.$$

This proves signed margin greater than one and $M(w) \le a + 1/8 \le 5a/4$.

The clone projections map this cylinder into itself. Row averaging preserves coordinate intervals because teacher rows agree within populations; right projection kills the bulk. The first-layer and readout intervals are likewise preserved. The merged network has the same strict signed margin, so both clipped outputs equal $f/2$. Moreover

$$\sqrt{E_v(w)} \ge a\sqrt{C/2} - \rho \ge (5/8)a\sqrt{C},$$

which gives $(C-1)M(w)^2 \le 4E_v(w)$. Thus the cylinder lies in $\mathcal{A}_N$.

For $Z \sim N(0, t^2)$ and any center m, direct integration of the Gaussian density gives

$$\mathbb{P}\{|Z - m| \le \rho t\} \ge \frac{2\rho}{e\sqrt{2\pi}} e^{-m^2/t^2} \qquad (\rho \le 1/16).$$

Multiply over the constrained coordinates and include the bulk probability 1/2. Since $t \le 1$ and $N/t^2 \ge 1$, the result is at least e^{-ND_*/t^2}. □

The radius decreases exponentially with depth. Only its logarithm enters the mass budget. The N^2 Gaussian bulk coordinates are never forced into a small Euclidean ball.

5 Compressed coordinates and finite coding

For fixed populations put

$$B^1 = F_1^\top W^1/\sqrt{N}, \quad B^s = F_s^\top W^s F_{s-1}, \quad b = \sqrt{N}F_H^\top v.$$

The merged network is exactly

$$g^1 = \mathrm{ReLU}(B^1\bar{x}), \quad g^s = \mathrm{ReLU}(B^s g^{s-1}), \quad z_{\bar{w}} = b^\top g^H.$$

Its full hidden state is $\sqrt{N}F_s g^s$; disjoint nonnegative columns and positive homogeneity let ReLU commute with this lift. No group-size coordinates are needed.

The entries of $\sqrt{d}B^1$, the internal B^s, and b are independent Gaussians of variance t^2/N. Set

$$\kappa = d\left\|B^1\right\|_F^2 + \sum_{s=2}^{H}\|B^s\|_F^2 + \|b\|_2^2, \quad p = q_1(d+1) + \sum_{s=2}^{H} q_s q_{s-1} + q_H.$$

Then

$$\mathbb{E}e^{N\kappa/(4t^2)} = 2^{p/2}, \qquad p/N \le d + 2 + (H-1)Q \quad (q_s \le \min\{N, Q\}). \tag{5}$$

Proof of Theorem 1. The target cylinder supplies a lower bound e^{-ND_*/t^2} on every noiseless posterior normalizer relative to the raw Gaussian, with either support convention. Since NE_v/t^2 is chi-square with N degrees of freedom,

$$\mu_{N,t}(E_v > B) \le e^{-NB/(8t^2)} \qquad (B \ge 8).$$

Its posterior probability is at most $\varepsilon/4$. A nonempty training set forces $M > 1/2$, so (1) implies $q < 1 + 16E_v$. Hence witnessing population counts on the retained posterior are at most Q.

There are at most $Q^{H(N+1)}$ tuples of partitions with counts at most $\min\{N, Q\}$. By (5), the raw probability that any tuple has $\kappa > J$ is at most

$$\exp\left[H(N+1)\log Q + \frac{N}{2}\{d + 2 + (H-1)Q\}\log 2 - \frac{NJ}{4t^2}\right].$$

Using $t \le 1, N \ge 1$ and (3), this is at most $e^{-ND_*/t^2}(\varepsilon/4)^{N/t^2}$. Its posterior probability is at most $\varepsilon/4$. The union is taken before selecting a witness from the realized network.

Every remaining network has a merged representation with coefficients bounded by $\sqrt{J}$. Pad to width Q with zeros, and round coefficients to fixed grids with coordinate error at most ζ, preserving zero. Both original and rounded coefficients have magnitude at most R. Hidden-coordinate magnitudes are bounded by A_g^s. Rounding errors obey

$$e_1 \le A_g\zeta, \qquad e_s \le A_g e_{s-1} + \zeta A_g^s, \qquad e_s \le s\zeta A_g^s.$$

The readout error is at most $(H+1)\zeta A_g^{H+1} = 1/32$. Thus the grid class $\mathcal{G}$, independent of N, t, satisfies

$$\log|\mathcal{G}| \le P_Q \log(3 + 4R/\zeta), \qquad |T(z_w) - T(g)| \le 3/32.$$

The second bound combines merging and rounding. Training inputs have $fT(g) \ge 13/32$, while a tested failure $fz_w \le 1/4$ entails $fT(g) \le 11/32$.

For fixed g, if the set $\{x : f(x)T(g(x)) \le 11/32\}$ has probability greater than $\varepsilon/2$, a sample avoiding it has probability at most $e^{-n\varepsilon/2}$. The union bound and (4) give one event on which every retained network has failure probability at most $\varepsilon/2$. The discarded posterior mass contributes at most $\varepsilon/2$. The common grid class proves simultaneity over all widths and scales. □

Proof of Corollary 2. Prune wires not ancestral to the single output, retaining required constants. Fan-in two yields width $C' \le \min\{C, 2^H + 1\}$ at the same depth. Apply the theorem with C' and use $\log(2 + C') = O(L)$. For $S = 2 + d + L + \log(4/\varepsilon)$, the explicit formulas give, with universal constants changing between inequalities,

$$\log(1/\rho) \le AS^2, \quad D_* \le AC'S^3, \quad Q \le AC'S^3, \quad J \le AC'S^4.$$

Hence $P_Q \le A(C')^2S^7$. Since $R \le A\sqrt{C'}S^2$ and $A_g \le A(C')^{3/2}S^5$, pruning also gives

$$\log(3 + 4R/\zeta) \le AS^2.$$

Thus $\log|\mathcal{G}| \le A(C')^2S^9 \le AC^2S^9$, proving the bound. □

6 The same conclusion for GELU

Let $\phi(u) = u\Phi(u)$ be exact GELU, where Φ is the standard normal distribution function. Replace every hidden ReLU by ϕ, and define the prior event (1) using this network and its merged counterpart. The same structural condition gives a polynomial-depth theorem. Here the Boolean circuit's compilation to GELU has a counted factor-two width overhead.

For a width-C Boolean teacher let $D = 2C$, and replace the target-cylinder budgets (2) by

$$S_{\mathrm{G}} = 4 + 4\sqrt{100D} + 2b_L, \qquad D_{\mathrm{G},0} = 2(d+1) + 2H\sqrt{D},$$

$$\rho_{\mathrm{G}} = \min\left\{\frac{1}{16}, \frac{1}{64a\sqrt{100D}\, D_{\mathrm{G},0} S_{\mathrm{G}}^{H}}\right\}, \qquad q_{\mathrm{G},0} = d + 2 + (H-1)D,$$

$$B_{\mathrm{G},0} = d + 100(H-1)D + 100a^2 D, \qquad D_{\mathrm{G},*} = B_{\mathrm{G},0} + q_{\mathrm{G},0} \log \frac{e\sqrt{2\pi}}{2\rho_{\mathrm{G}}} + \log 2.$$

Use $D_{\mathrm{G},*}$ in place of D_* in the definitions of B, Q, J, R. Use instead

$$A_{\mathrm{G}} = 4(d+1+Q)(R+1), \qquad \zeta_{\mathrm{G}} = [32(H+1)A_{\mathrm{G}}^{H+1}]^{-1}, \qquad P_{\mathrm{G}} = P_Q + HQ.$$

Theorem 4 (GELU version)**.** *Theorem 1 holds for the GELU prior just defined, simultaneously for every $N \geq 2C$ and $0 < t \leq 1$, if its sample bound is replaced by*

$$n \geq \frac{2}{\varepsilon}\left[P_{\mathrm{G}} \log(3 + 4R/\zeta_{\mathrm{G}}) + \log(1/\delta)\right].$$

The optional all-input margin restriction is still allowed. The width dependence remains $O(C^2 \log^3(2+C))$ at fixed other parameters. Corollary 2 holds after changing its universal constant.

Proof. We give the two changes needed to the ReLU proof: an exact Boolean compiler and a uniformly well-conditioned family of compressed activations.

Exact Boolean compiler. Set $\tau = \Phi(1) > 2/3$, $s = 2\tau - 1$, $A = \tau/s$, and $B = (1-\tau)/s$. Then $1 < A < 2$ and $0 < B < 1$, and direct substitution gives

$$A\phi(u) + B\phi(-u) = \mathrm{ReLU}(u) \qquad (u \in \{-1, 0, 1\}).$$

Represent every logical bit, including every constant-one wire, by two GELU neurons with preactivations $u, -u$. Decode its bit by the displayed linear combination. For NOR, COPY, and NOT use respectively $u = 1 - b_1 - b_2$, $u = b_1$, and $u = 1 - b_1$. Decode the constant-one wire for the internal offset. First-layer input bits are $(1 + x_i)/2$. All preactivations lie in $\{-1, 0, 1\}$ on Boolean inputs, so the simulation is exact at the same depth and width $D = 2C$.

Every hidden coordinate has magnitude at most one. First-layer row squared norms are at most one. An internal row has squared norm at most $5(A^2 + B^2) < 25$, including repeated gate inputs. Before cloning, the readout for $af = a(2b_{\mathrm{out}} - 1)$ has squared coefficient norm between a^2 and $25a^2$: if output and constant coincide, combine their coefficients. Balanced cloning into width $N \geq D$ therefore supplies the conservative bounds

$$\|W_*^1\|_F^2 \leq N, \qquad \|W_*^s\|_F^2 \leq 100D, \qquad a^2 D/2 \leq N\|v_*\|_2^2 \leq 100a^2 D.$$

These exact teacher states are constant on the D clone populations.

GELU is globally 2-Lipschitz and vanishes at zero. The cylinder proof therefore has error recurrences

$$e_1 \leq 2\rho_{\mathrm{G}}(d+1), \qquad e_s \leq S_{\mathrm{G}} e_{s-1} + 2\rho_{\mathrm{G}}\sqrt{D}.$$

The chosen radius gives $e_H \le 1/(64a\sqrt{100D})$. The same readout and projection calculation proves signed output margin greater than one, clipped merging equality, and the normalized population-count budget. Its raw probability is at least $e^{-ND_{G,*}/t^2}$. Although the approximate network is not an exact digital circuit, its errors are controlled by the chosen parameter radius; digital restoration is not needed.

Population masses in the compressed model. For a population of size k, put $r = \sqrt{k/N} \in (0, 1]$ and define

$$\psi_r(u) = r\phi(u/r), \qquad \psi_0(u) = \mathrm{ReLU}(u).$$

Writing φ for the standard normal density, differentiation for $r > 0$ gives

$$\partial_r \psi_r(u) = \phi(u/r) - (u/r)\phi'(u/r) = -(u/r)^2\varphi(u/r), \qquad |\partial_r \psi_r(u)| \le \frac{2}{e\sqrt{2\pi}} < 1.$$

Also $|\phi'| \le 2$. The extension at $r = 0$ is continuous uniformly in u, since $\sup_u |\phi(u) - \mathrm{ReLU}(u)| < \infty$. Consequently

$$|\psi_r(u) - \psi_{r'}(u')| \le 2|u - u'| + |r - r'|, \qquad |\psi_r(u)| \le |u| \quad (0 \le r, r' \le 1). \tag{6}$$

The exact normalized merged computation is now

$$g_j^1 = \psi_{r_{1,j}}((B^1\bar{x})_j), \qquad g_j^s = \psi_{r_{s,j}}((B^s g^{s-1})_j), \qquad z = b^\top g^H, \quad r_{s,j} = \sqrt{k_{s,j}/N}.$$

The Gaussian distribution of $\sqrt{d}B^1, B^s, b$ is unchanged. Thus the same readout and partition-union bounds retain a posterior of mass at least $1 - \varepsilon/2$ with $q_s \le Q$ and coefficient norm budget J.

For finite coding, append at most HQ population-mass parameters in $[0, 1]$, including zero masses for padded zero neurons. Round these and all coefficients with coordinate error at most ζ_G. Equation (6) and the enlarged A_G give hidden magnitudes at most A_G^s and rounding errors at most $s\zeta_G A_G^s$. The output error is at most $1/32$. The grid has log-cardinality at most $P_G \log(3 + 4R/\zeta_G)$. Its elements are auxiliary real-valued predictors; they need not themselves be width-N networks or have realizable population proportions. The finite-class argument uses only their predictions and therefore applies unchanged. The same ancestral pruning and estimates as in Corollary 2 give the stated literal polynomial. □

7 Interpretation and limits

If $p(x) = \mathbb{P}_w\{f(x)z_w(x) > 1/4\}$, then the theorem implies

$$\mathbb{P}_X\{p(X) \le \theta\} \le \varepsilon/(1 - \theta), \qquad 0 < \theta < 1.$$

The strict $> 2/3$ criterion follows by adjusting accuracy by a factor of three. Correct-sign probability is at least the correct-margin probability. No MAP predictor is silently substituted for the posterior.

The restriction expresses stability under a specific structured change of the weights. It is not proved equivalent to robustness against arbitrary calibrated internal disturbances, nor established for typical trained networks. Its positive-mass Boolean implementation shows useful representability, not universal empirical applicability. An arbitrary analog ReLU teacher is not identified with a Boolean circuit of the same width and depth.

These theorems therefore give polynomial-depth results for specified restricted priors, at all mean-field-or-stronger scales. They do not settle the unrestricted polynomial-depth learning question. Exponentially small parameter radii are harmless in this construction because only their logarithm is paid; exponentially large compressed width would not be harmless, and the population condition prevents that loss.

Provenance and verification. The proof adapts the earlier collaborative tanh population construction and Gaussian partition bound, including contributions from Claude. The ReLU normalized representation, GELU perspective extension and exact compiler, noncontracting target-cylinder estimate, and resulting depth-polynomial theorems are supplied here. This research manuscript is not Lean verified.